Game Design Document (GDD)

Project Codename: Trion Tactics (World Trigger Battle Sim)

1. Executive Summary & Vision

- **Genre:** Cross-Platform Online Turn-Based Strategy (Simultaneous Execution / Asynchronous)
- **Target Platform:** Primary: Mobile (iOS / Android) | Secondary: PC
- **Engine:** Godot Engine 4.x
- **Core Inspiration:** *World Trigger* Away Mission Test Battle Simulation Exercise
- **Visual Style:** 3D Low-Poly "Hologram Tabletop" (Chess-like stylized diorama board with orthographic/isometric view and minimal animations)

Design Pillars

1. **Low Floor, High Skill Ceiling:** Easy-to-grasp card-to-timeline dragging on mobile, backed by deep multi-layered systems (Dual-Slot interference, Penchants, simultaneous reaction counters, and spatial control).
2. **Predictive Simultaneous Execution:** Tactical focus shifts from direct twitch reflexes to predicting enemy movement, timing parries, and reading combat timelines across 6 discrete ticks.
3. **Asymmetric Deck Building:** Pre-match squad customization using an 8-slot Trigger Holder constrained by a global Trion budget.

2. Platform, Scope & Technical Architecture

2.1 Target Platforms & UX Strategy

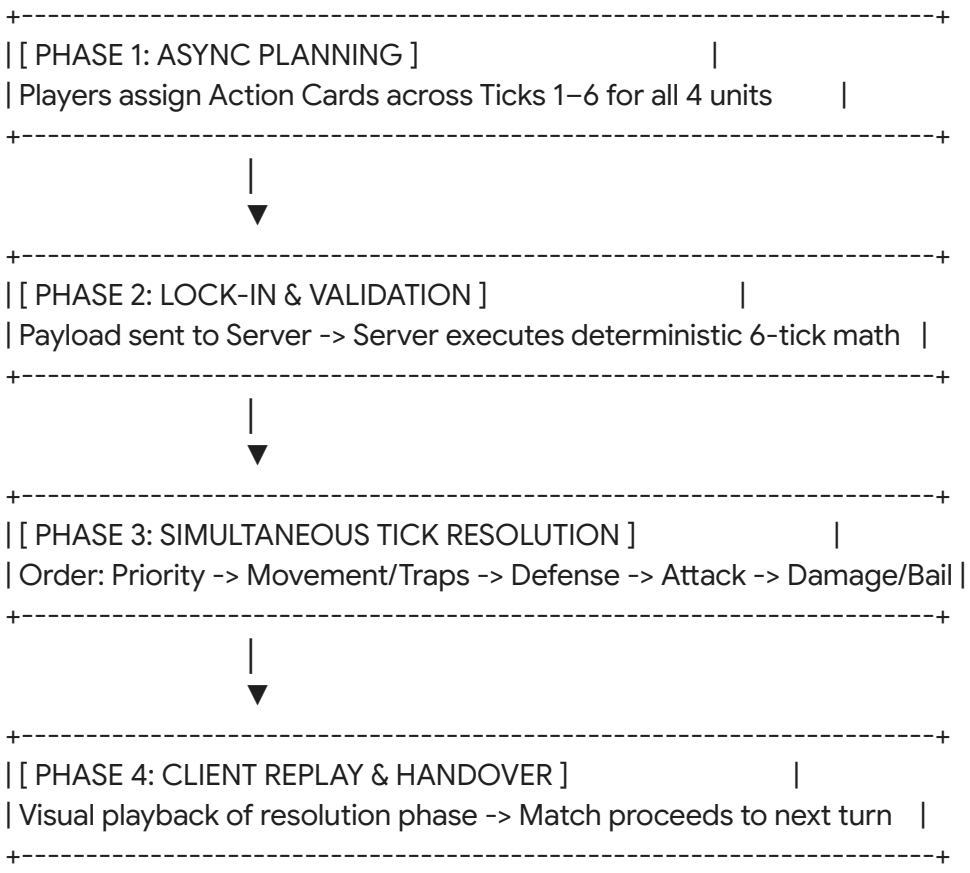
- **Mobile-First Paradigm:** Touch-friendly UI with prominent hitboxes, swipeable top squad-selection bar, and bottom drag-and-drop card interaction.
- **Cross-Platform PC Support:** Unified input mapping; mouse clicks replace tap gestures, and scroll wheel controls pinch-zoom on the 3D diorama map.

2.2 Network Architecture

- **Hybrid Networking Framework:**
  - **HTTP / REST API Backend:** Handles asynchronous match state, database persistence, turn submissions, loadout validations, and rating/matchmaking.
  - **Server-Authoritative Execution Engine:** A headless Godot server instance (or custom lightweight server logic) processes turn data deterministically upon both players locking in (or timer expiration).
  - **Built-in Godot RPC System (@rpc):** Leveraged for real-time spectating, live replay playback, or direct low-latency synchronization when both players are online concurrently.

3. Core Match Structure & Turn Lifecycle

Matches play out in 4v4 Squad Skirmishes over a series of synchronous execution turns. Each turn is subdivided into **6 Discrete Ticks**.



3.1 Time Limit & Penalty Enforcement

- **Turn Limit:** Asynchronous turn timer (e.g., 12 to 24 hours per turn).
- **Timeout Resolution:** If a player fails to submit before the timer expires, the server auto-populates the unsubmitted units' queues with **"Stand Ground + Set Main Shield"** defensive fallbacks.

4. Grid, Board Geometry & Line-of-Sight

4.1 Hexagonal Grid Coordinates

- Uses an **Axial Coordinate System** (q, r).
- All 6 directional transitions require equal movement expenditure, eliminating diagonal anomalies.

4.2 Shielding, Cover, and Flanking

- **Hex-Edge Shielding:** Shields cover 3 specific contiguous edges (e.g., *Front-Left*, *Front*,

Front-Right).

- **Flanking Flaw:** Attacks entering from the 3 unshielded rear edges bypass defense calculations completely unless the unit is in **Full Guard**.
- **Cover Types:**
  - **Half Cover (50%):** Reduces incoming raw attack damage by 50%.
  - **Full Cover (100%):** Blocks standard ballistic trajectories completely.
  - **Curved Trajectories (Viper):** Allows players to route bullet paths around hex cover to strike exposed rear edges.

4.3 Radar, Stealth, and Line-of-Sight (LoS)

Vision State	Trigger Condition	Map Presentation	Data Revealed
Radar Signal	Default / Out of LoS	Ghost Ping offset by 1–2 hexes randomly	Presence of unit only (No identity, HP, or facing)
Direct Sight	Clear Raycast to Unit	Full 3D Low-Poly Model on exact hex	Unit ID, HP (Trion), Facing direction, Active Triggers
Bagworm Stealth	Bagworm Trigger Active	Completely removed from enemy Radar	Hidden until enemy enters within 3 hexes or unit attacks

5. In-Match Resource & Action Mechanics

5.1 Action Tokens (AT) Economy

- Each unit receives a base pool of **Action Tokens (AT)** per turn (Standard base: $8AT$).
- Actions consume AT from the pool during the planning phase. Unused AT does not carry over to subsequent turns.

5.2 Basic AT Expenditure Matrix

- **Movement:** $1AT$ per hex moved.
- **Standard Attack (Single Slot):** $3AT$.
- **Standard Shield (Single Slot):** $2AT$ (Mitigates damage equal to Defense stat).
- **Bagworm (Stealth):** $1AT$ maintenance cost per turn.

5.3 Advanced High-Skill Ceiling Mechanics

Dual-Slot Interference (Main / Sub Slots)

- Each tick features **Main** and **Sub** action slots.
- Running concurrent actions on both slots during the same tick incurs a $+\ 1AT$ **Mental**

Load Penalty.

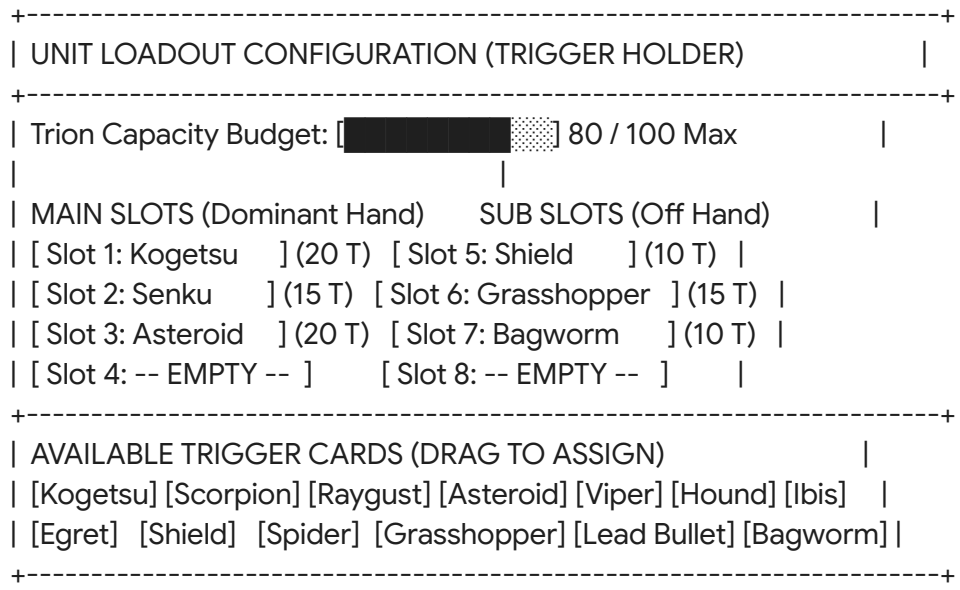
- **Full Guard:** Activating *Main Shield + Sub Shield* simultaneously consumes $4AT$, granting **$+50\%$ Defense**, but completely locking out movement and offensive actions for that tick.

Penchants & Synergy Passives

- **Kogetsu / Scorpion Penchant:** Equipping primary melee triggers increases base AT pool to $10AT$.
- **Wire Mobility Penchant:** Units standing in or moving across hexes containing **Spider Wires** gain $+20\%$ Movement range, $+20\%$ Defense, and $+10\%$ Evasion.
- **Close-Quarters Combo (Melee Assist):** If Unit A executes a melee strike on a target adjacent to friendly Unit B, Unit B performs an automatic follow-up attack at $0AT$ cost.
- **Ranged Combo (Crossfire Assist):** If a ranged attack targets an enemy situated inside an aligned friendly unit's firing arc, the ally performs a follow-up attack at 50% reduced AT cost.

6. Pre-Match Deck Building: Trigger Holder Customization

Before queuing for matches, players configure each of their 4 squad units in the **Hangar Screen**.



6.1 Constraints & Rules

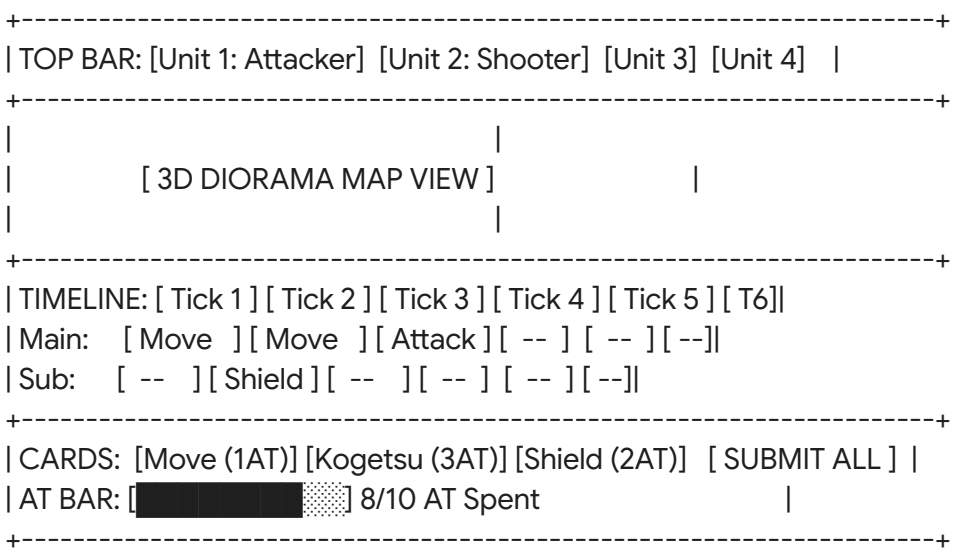
1. **8-Slot Capacity:** 4 Main Slots + 4 Sub Slots.
2. **Trion Capacity Budget:** Every equipped Trigger Card adds to the unit's total Trion

loadout cost. The total sum cannot exceed the unit's **Max Trion Capacity (100)**.

3. **Card Deck Generation:** Equipped Trigger Chips dictate the available **Action Cards** present in the unit's in-battle timeline deck.

7. User Interface (UI) & User Experience (UX)

7.1 Battle Screen Layout



7.2 Core Interaction Loops

1. **Unit Quick-Select:** Tapping a unit portrait in the top bar re-centers the orthographic 3D camera onto that unit and loads their timeline.
2. **Drag & Drop Timeline Assembly:** Dragging an Action Card into a tick slot assigns the command. Dropping cards onto reachable map hexes sets target destination or attack directions.
3. **Multi-Unit Lock-In:** The global **"Submit All"** button activates only after all 4 units have valid timeline configurations without AT overruns.

8. Technical Architecture in Godot 4

8.1 Key Data Structures

TriggerCard.gd (Resource)

```
class_name TriggerCard
extends Resource

@export var card_name: String
@export var trion_equip_cost: int     # Deck-building budget cost
@export var action_at_cost: int      # Timeline placement cost
@export var is_dual_slot_compatible: bool
@export var card_icon: Texture2D
```

UnitData.gd (Resource)

```
class_name UnitData
extends Resource

@export var unit_name: String
@export var base_trion_hp: int
@export var base_at_pool: int
@export var mobility_rating: int
@export var attack_rating: int
@export var defense_rating: int

@export var main_trigger_slots: Array[TriggerCard] = [] # Max 4
@export var sub_trigger_slots: Array[TriggerCard] = []  # Max 4
```

TickAction.gd (Object / Struct)

```
class_name TickAction
extends RefCounted

var tick_index: int          # 1 to 6
var main_card: TriggerCard
var sub_card: TriggerCard
var target_hex: Vector2i      # Axial (q, r)
```

8.2 Server Tick Resolution Pipeline

The deterministic server engine processes the 6 ticks sequentially during Phase 3 using the following sub-phase priorities:

Sub-Tick Priority Execution Order:
1. State & Priority Phase -> Toggles (Bagworm ON/OFF, Stance Shifts)
2. Traps & Movement Phase -> Spider Wire Deployments, Hex Position Shifts
3. Defensive Phase -> Shield / Full Guard Positioning
4. Attack & Projectile -> Melee Strikes, Ranged Bullet Trajectories
5. Resolution Phase -> Hit/Evasion Rolls, Trion (HP) Deduction, Bail-Outs

9. Development Roadmap & Phased Implementation

Phase 1: Core Mechanics & Prototype (Local 1v1)

- Implement hex grid math, axial coordinate conversions, and pathfinding in Godot 4.
- Build the 6-tick timeline ribbon and drag-and-drop UI controls.
- Implement deterministic 6-tick local turn execution for 2 units.

Phase 2: Complete Combat Mechanics & Deck Customization

- Integrate 4v4 unit switching and squad management UI.
- Add line-of-sight raycasting, cover mitigation, and hex-edge shielding calculations.
- Implement Radar pings and Bagworm stealth mechanics.
- Build the Hangar Deck Building screen with Trion budget validation.

Phase 3: Backend & Asynchronous Multiplayer

- Set up database models for user accounts, squad loadouts, and match states.
- Implement HTTP turn queue submission and validation API.
- Implement deterministic server-side resolution runner.
- Add turn playback / replay visualization on the mobile client.

Phase 4: Polish, Cross-Platform UI & Balancing

- Refine 3D low-poly assets, shaders, hex highlights, and toon outlines.
- Optimize touch targets and responsive UI scaling for mobile/tablet aspect ratios.
- Balance AT costs, trigger Trion weights, and penchant modifiers.

Next Steps

Now that the complete GDD is established, we can begin building out individual systems—such as writing the core Godot hex math scripts, prototyping the UI nodes, or refining specific trigger card stats! How would you like to proceed?